higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

COMPUTER HARDWARE AND SOFTWARE
NQF LEVEL 3

12 NOVEMBER 2019

This marking guideline consists of 8 pages.

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SECTION A

QUESTION 1

1.1 True
1.2 False
1.3 True
1.4 True
1.5 True
1.6 True
1.7 True
1.8 False
1.9 True
1.10 False

(10 × 1) **[10]**

QUESTION 2

2.1 C
2.2 A
2.3 B
2.4 G
2.5 F
2.6 D
2.7 E
2.8 M
2.9 H
2.10 L

(10 × 1) **[10]**

QUESTION 3

3.1 C
3.2 A
3.3 B
3.4 C
3.5 B
3.6 A
3.7 C
3.8 B
3.9 D
3.10 C

(10 × 1) **[10]**

TOTAL SECTION A: 30

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SECTION B

QUESTION 4

4.1

- System software – this is a type of computer program that is designed to run a computer's hardware and application programs.
- Application software – is any program, or group of programs, that is designed for a specific task for the end user.
- Kernel – is the central part of an operating system that manages the operations of the computer and the hardware, most notably memory and CPU time.
- Assembler – is a program that takes basic computer instructions and converts them into a pattern of bits that the computer's processor can use to perform its basic operations.
- Firmware – software instructions residing in non-volatile storage that holds its content without power. Firmware is found on computer motherboards to hold hardware settings and boot data (see BIOS) and on myriad consumer electronics devices to hold the operating system.
- Hardware – includes the physical parts or components of a computer, such as the central processing unit, monitor, keyboard, computer data storage, graphics card, sound card, speakers and motherboard. (Any 5 × 2)

4.2

- Static RAM (SRAM)✓ – Used for cache memory and PCMCIA (Portable Computer Memory Card Industry Association) memory cards.✓
- Dynamic RAM (DRAM)✓ – Most commonly used for primary or main memory on a PC. It is commonly referred to as simply RAM.✓
- Parameter RAM (PRAM)✓ – Used on Macintosh computers to store internal information, such as the computer's date and time and other configuration data that must remain in memory after the computer powers down.✓
- Pseudo-static RAM (PSRAM)✓– Specifically made for use in portable computers.✓
- Video RAM (VRAM)✓– Used on video adapter cards for buffering between the PC system and the video display.✓ (5 × 2)

(10 × 2) **[20]**

QUESTION 5

5.1
- Memory management: Kernel provides virtual and physical memory to processes for completing their execution.
- Device management: Kernel controls the activity of peripheral devices through device drivers.
- Scheduler: Kernel acts as a scheduler for processes during execution. At a given time, a single process can be executed by the processor. (Any 2 × 2) (4)

5.2
- How far is the installation site from the office?
- Is there parking close to the installation office?
- Is the installation site on the ground floor or upstairs in a multi-storey building?
- Does the equipment have to be carried upstairs, if the site is a multi-storey?
- Is someone available to help you carry the equipment or boxes? (5)

5.3
- If it's an external storage device like USB, plug it into the port and confirm that the computer recognises the device.
- If it's a printer, print a document on the printer.
- If it's a router, connect to the internet.
- If it's a scanner, scan and confirm that the document is accessible on the computer.
- If it's a backup device, make a backup and then restore data. (5)

5.4 Multifunction device (1)

[15]

QUESTION 6

6.1
- Anti-static wrist strap
- Multimeter
- Torch
- Compressed air
- Needle nose or long nose pliers
- Flat blade screw drivers
- Phillips screwdrivers (Any 5 × 1) (5)

6.2
- A bottle of 70% isopropyl alcohol (a quarter bottle is more than enough).
- A can or two of compressed air.
- A package of non-shredding cleaning tissues, a soft lint-free cloth, or a Scotch-Brite HPEC cloth. The Scotch-Brite cloth is recommended.
- A package of high-quality cotton swabs.
- A Chinese bristle artist's brush, which has bristles about two inches long and is one inch in diameter.
- An inexpensive pair of pointed-tip tweezers.
- A small brush-head vacuum cleaner for cleaning the keyboard and inside the system case.
- A medium-size Phillips screwdriver for case, keyboard, and adapter board screws.
- A bottle of non-ammonia window cleaner to clean the glass on the monitor.
- An ESD grounding strap. You can also get ESD mats to place the PC on.

(Any 7 × 1) (7)

6.3
- Back up the system – Anytime you open the system case to add, remove, or replace components such as the processor, memory, the power supply, or a disk drive, you should create a backup of the hard disk drive, especially if you are working on the hard disk drive itself.
- Protect against ESD – Always protect the PC against ESD (electrostatic discharge), the static electricity that can build up in the PC and you.
- Work in a well-lit area – Most of the components in the PC are small, especially the screws. Anytime you open the system case, you should do so in a work area with plenty of direct light.
- Protect the memory module – Keep all memory modules in their protective packaging right up to the moment you are ready to install them.
- Handle modules only by their edges – Avoid touching the module's connectors and components. (5)

[17]

QUESTION 7

7.1
- Speed of hardware-software communication.
- Stability of hardware, software and network.
- Accuracy of the results and readings on tested processed data.
- Communication between workstations and file servers on a network.
- User friendly, for it to be accessed by randomly selected people.
- Integrity in determining how well the system maintains relationship between processed data.
- Security for protection against viruses, spyware and hackers.

(Any 6 × 1)

7.2
- Lack of personnel when testing is to begin.
- Lack of required hardware, software, data or tools.
- Late delivery of the software and hardware tools.
- Delays in training testers on the application or on any tools to be used.
- Changes to the original requirements or designs.
- Complexes.

(2 × 6) **[12]**

QUESTION 8

8.1 End User Licence Agreement (2)

8.2
- The EULA licenses a software program to the user but doesn't sell the app to the user✓ – Providing a license to your users that allows them to use your program according to your limits and restrictions rather than providing them with full ownership of the app, allows you to maintain far more control over the distribution and use of your app.✓
- An EULA restricts certain abusive uses of your app✓ – One of the main advantages to any sort of legal agreement, such as a Terms and Conditions or EULA, is that you can use these agreements to restrict undesirable uses of your app.✓
- An EULA allows you to limit your liability✓ – As a software developer, limiting your liability is very important. If you don't do so, you're potentially opening yourself and your business up to lawsuits, many of which may be frivolous but still time consuming and financially burdening.✓
- An EULA allows you to disclaim warranties.✓ Disclaimer of Warranties clause in your EULA allows you to provide your app licence and app on an "As Is" and "As Available" basis, and places the general risk and discretion of use on your users.✓
- An EULA can protect your right to terminate licenses or stop operating without repercussion✓ – While an EULA is implemented for granting licences, these agreements are also convenient for maintaining your right to revoke licences.✓

(Any 4 × 2) (8)
[10]

QUESTION 9

9.1
1. First, check the printer to see that it is turned on and plugged in to the surge protector. If it is, that is not the issue.
2. Next, check the printer's USB cord and find out if it is plugged in or not.
3. Next, you check to make sure the printer's ink cartridge still has ink. If it does have, that is not the issue.
4. Thereafter you check that there is paper loaded in the paper tray. If there is, you know the issue has nothing to do with ink or paper.
5. Now you want to make sure the printer and computer are communicating correctly.
6. You check the printer's USB cord and find that it is not plugged in.
7. Check if there is any recently downloaded update of an operating system, as it might interfere with the printer. (Any 5 × 1)

9.2
- Report on all symptoms found.
- Report on all the troubleshooting techniques used.
- Report on all the results of troubleshooting.
- Report on the knowledge bases where a search for solutions was done.
- Report on technical manuals consulted.

(2 × 5) **[10]**

QUESTION 10

10.1
- Users – people usually make the most mistakes when they use computers, so their actions are often the cause of problems.
- Applications – If a computer has technical problems; one application may be causing them.
- Operating system – if users and applications are not causing the problem, it could be the operating system.
- BIOS – the BIOS might be causing problems with the way in which the hardware and software communicate.
- Hardware – If there are no problems with the rest of the above-mentioned, the computer could have a hardware fault. (5)

10.2
- Prioritise the problem✓ – assign a priority to the call and make a note of what promises you have made.✓
- Identify the user and their terms of support so the response procedure can be determined ✓– not all users have the same support contract, some have full support contracts and some have no contract at all.✓
- Record sufficient information about the problem to begin an investigation.✓ Get as much information about the problem as you possibly can. If it is a new problem, take responsibility and ensure that you get all relevant information to solve the problem.✓
- Employ a personal communication technique with users✓ – inform the user how you intend communicating progress on solving the problem, e.g. use of e-mails or phones.✓
- Assign a time frame to the problem✓– communicate the response time to be expected in order to resolve the problem to the caller or user.✓ (5 × 2) (10)

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QUESTION 11

11.1
- BIOS ROM checksum error – This suggests that the data stored inside the BIOS has been rendered invalid or inaccessible and you need to contact the motherboard or BIOS manufacturer for recovery procedures, if any.✓
- CMOS battery failure✓ – The CMOS battery is dead and needs to be replaced.
- CMOS checksum error✓ – defaults loaded. This message indicates that the CMOS has become corrupt and the setup configuration data has been reset to the default values.✓
- Display switch is set incorrectly✓ – Many older motherboards have a jumper that sets the type of video display being supported. If this error displays, the video jumper setting and the video configuration in CMOS are in conflict.✓
- Floppy disk failure✓ – The BIOS POST is looking for a floppy disk controller (FDC) that is indicated in the CMOS.✓
- Hard disk failure✓ – The BIOS POST cannot find or initialise a hard disk controller (HDC) included in the CMOS configuration.✓ (Any 5 × 2) (10)

11.2
- Were other people in the office at the time?
- Was equipment bumped or knocked over?
- Did something get spilled on the equipment?
- Was equipment moved recently?
- Was other mechanical or electrical equipment in use close by?
- Has there been recent thunderstorm activity? (Any 5 × 1) (5)

11.3
- Automatic document feeder
- Duplex printing
- Multiple paper sizes
- Output to email
- Output to internet storage
- Retrieval from internal storage (6)

[21]

TOTAL SECTION B: 120
GRAND TOTAL: 150

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